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DEM simulation of solid-electrolyte separator and cathode densification using a stress-based multi-contact elasto-plastic model [☆]Cerun Alex Varkey ^{a,b,*}, Kostas Giannis ^b, Sebastian Melzig ^{a,b}, Carsten Schilde ^b, Sabrina Zellmer ^{a,b}^a Fraunhofer Institute for Surface Engineering and Thin Films IST, Riedenkamp 2, 38108 Braunschweig, Germany^b Institute for Particle Technology (iPAT), Technische Universität Braunschweig, Volkmaroder Straße 5, 38104 Braunschweig, Germany

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ABSTRACT

Advancements in high-performance all-solid-state batteries (ASSBs) rely on achieving high relative densities of separators and cathodes to enable efficient ionic and electronic transport. Integrating experiments with DEM simulations provides a particle-scale analysis of microstructure evolution, contact-network formation, and force response. Predicting densification via DEM is challenging because conventional DEM treats particles as rigid bodies with soft local contacts, which becomes inaccurate as porosity decreases and contact networks become highly constrained. Moreover, capturing the coupled effects of high plasticity and evolving contact area during compaction remains difficult. In this work, we introduce a modeling framework to simulate densification of ASSB cathodes and separators by integrating an elasto-plastic contact model with plastic contact-deformation approaches that overcome local-contact assumptions through non-local force calculations using stress-based multi-contact formulations. The model is compared with the Hertz contact model, Thornton–Ning elasto-plastic model, and experimental data for both unimodal and bimodal PSDs. Simulations at higher load levels show that our contact model captures stronger nonlinearity in higher stress regions and improves agreement with experiments in the dense regime. This approach enables predicting different compaction pressures during cell assembly and cycling, helping to identify suitable processing parameters that enhance the electrochemical properties of the components.

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1. Introduction

All-solid-state batteries (ASSB) are electrochemical storage devices that replaces the liquid or gel electrolyte of the conventional Li-ion batteries (LIB) with a solid electrolyte facilitating improved ionic conductivity and interfacial stability. They are of increasing research interest because of its high energy density and improved safety by eliminating flammable liquid electrolytes [1,2]. Accordingly, the development of modeling frameworks to simulate the densification process of ASSB components has gained significant attention, emphasizing the importance of accurately capturing stress distribution, deformation behavior, and interparticle interactions, all of which strongly influence microstructural integrity and electrochemical performance.

Solid electrolytes are broadly classified into oxides, halides, sulfides and polymer systems, with each electrolyte systems having

different advantages and disadvantages in terms of ionic conductivity, processability, mechanical integrity and interfacial stability [2–5], among which halide-based system exhibit a combination of high ionic conductivity, low-temperature processability and superior chemical stability for scalable cell fabrication [3,6]. Halide solid electrolyte (Li₃YBrCl₅) exhibit high Li⁺ ion conductivity, high deformability and high electrochemical stability [6]. They also exhibit stable cycling in a cell system with Li metal anodes and cathodes with nickel-rich cathode active materials (CAM), such as NMC-811 (LiNi_{0.8}Mn_{0.1}Co_{0.1}O₂) [7].

An ASSB cell consists of a negative electrode, typically a Li metal electrode, a separator layer with solid electrolyte particles and a positive electrode [2]. The positive solid-state electrode consists of cathode active material (CAM), solid electrolyte (named catholyte), a conductive agent and a binder [8]. In ASSB, ion transport occurs via thermally activated hopping through crystalline or amorphous solid electrolytes, strongly influenced by defects and interfacial resistance [2]. In contrast, LIB rely on solvated ion diffusion in liquid electrolytes, enabling higher mobility but poorer mechanical and thermal stability [5]. Electron transport in composite cathodes is primarily mediated through the conductive

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agent network, while Li^+ ion conduction depends on the continuous solid-electrolyte network and is hindered in CAM-rich regions that impede ionic pathways [9]. This requires a well-coordinated interplay between an electron-conducting network with a high density of conductive pathways and an equally well-developed electrolyte network that ensures efficient ionic transport. Consequently, an accurate representation of the binder network is a key factor to consider while modelling the electrode structure. Fig. 1 illustrates the particulate structure of the separator and cathode in a solid-state battery, also providing an insight into the e^- and Li^+ ion transport mechanisms.

Densification of the solid-state cathode and separator films is a crucial step in the manufacturing of ASSB due to its significant influence on the microstructure of the cathode and the separator and the overall performance of the battery. During densification, applied pressure drives successive microstructural changes including particle rearrangement, elastic and plastic deformation at interparticle contacts resulting in pore reduction and increasing interparticle contacts critical for ionic and electronic transport [2]. Understanding how to adjust the degree of densification is important for improving the e^- and Li^+ ion conduction and mechanical properties of the electrodes [10]. Development of high performance ASSB relies heavily on achieving high relative densities to ensure sufficient ionic transport and reduced interfacial resistance [10]. Unlike conventional LIBs, where porosity is essential for the Li^+ migration, the ASSB components should be densified to achieve very high relative densities (> 95–98%), as residual porosity reduces ionic conductivity and degrades electrochemical performance [11]. However, this densification target must be understood in the context of a complex interplay between ionic transport, mechanical integrity and fracture behavior. While relative high densities generally improve the ionic percolation through the solid electrolyte network and reduce grain boundary resistance, extensive compaction can induce particle fracture in CAM particles, disrupting the electronic conduction network [12,13]. Further, a complete elimination of pores is unrealistic in practice and has a negative impact on the mechanical stress relaxation within the ASSB components [2]. Also, mechanically robust solid electrolytes are necessary for long term stability of the battery. Key mechanical metrics, such as elastic modulus, strength, and toughness, are critical for assessing the load-bearing capacity, deformation characteristics, and failure modes of solid electrolytes in ASSBs [14–16].

1.1. DEM modeling of ASSB

In the electrode manufacturing process, densification is one of the key steps to reduce porosity and enhance inter-particle contacts [17]. When higher pressure is applied during fabrication, the particles are forced into closer proximity, reducing the void space and increasing the number of particle contacts. In particle packing studies, it is shown that the coordination number of the particles increases with increasing packing fraction [18]. For random loose packing of monodisperse spheres, packing fractions of approximately 0.59–0.60 are observed with a mean coordination number of 4–6, while random close packing yields packing fractions of 0.64 with coordination numbers of 6–8, typically referred as dense packing [19,20]. This gives rise to a complex mechanical system in which particles are no longer governed by binary collisions. Simulating the densification process using DEM poses a significant challenge, as conventional DEM approaches based on the soft-sphere model – which treats particles as rigid bodies with overlapping, deformable contacts where contact forces are calculated from the degree of overlap, fail to yield physically meaningful results when porosity decreases. This is associated with the inaccurate representation of the significant deformation and multi-body interaction that occur in a highly dense granular media [21]. To address this issue, new contact formulations are studied in order to capture complex physics of soft, deformable materials under high stress [22,23].

In classical DEM, the formulation considers independent contact assumption, which means each interacting particle pair is treated independently and isolated from other contacts. Local calculation of the contact forces neglects the influence of stress redistribution in the particles volume. Different approaches are suggested in the literature to address this limitation of the DEM framework. Brodu et al. proposed a model to consider the mutual influence of contacts that are acting on a particle by considering the overall deformation of the particle using a strain field [21]. Giannis et al. defined a different approach for the multi contact DEM in which the trace of the particle's stress tensor, which incorporates information from all local contacts, thereby enabling the model to account for anisotropic particle deformation [24].

Standard Hertzian contact models, which assume purely elastic behavior, are insufficient for predicting the final density and stress distribution of low porous ASSB electrodes [25]. To answer this problem, elastoplastic contact laws are established. The Edinburgh

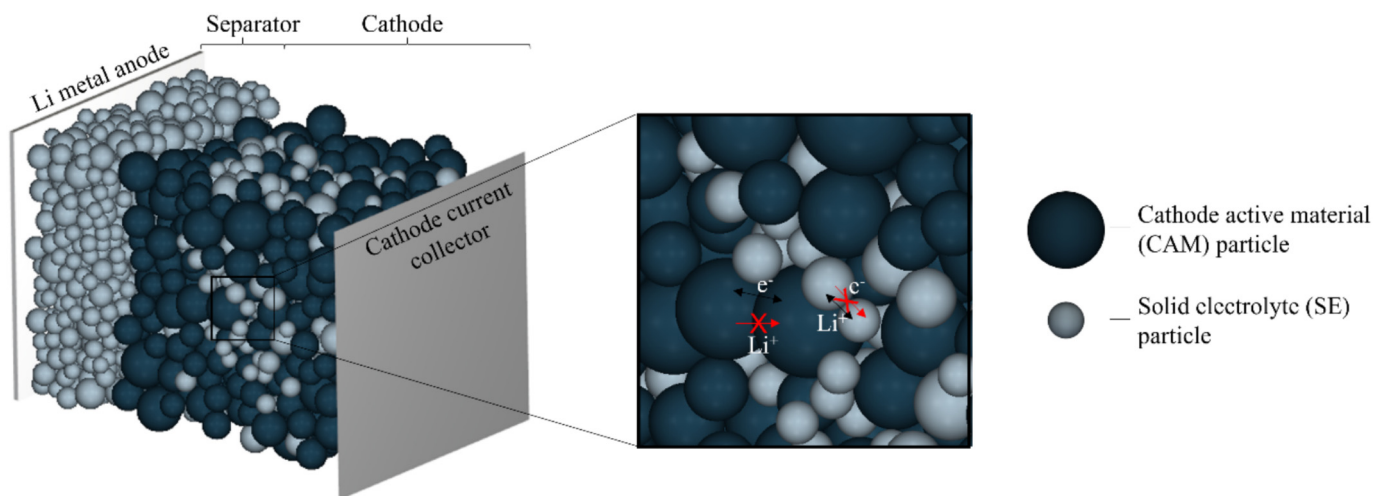


Fig. 1. Particulate structure of a solid-state separator and integrated cathode as well as depiction of the diffusion mechanisms in an ASSB cell.

Elasto-Plastic Adhesion (EEPA) model introduces a hysteresis loop in the force–displacement relationship, which allows for the dissipation of energy through the plastic deformation [26]. So et al. developed a discrete element method that can simulate the plastic deformation of particles, focusing on sintering and electrode compaction of all-solid-state batteries. The model can simulate mold compaction processes, differentiating the elastic, plastic, and viscoelastic deformations regimes. Inelastic deformation occurs when the stress exceeds the yield strength of the material, which is then described by either a plastic or viscoelastic response [25]. The Thornton and Ning model derives the force–displacement relationships and divides it into distinct phases based on the contact pressure. At small overlaps, the deformation is reversible and elastic. In this phase, the force–displacement relationship follows the Hertzian contact model. The model defines a yield point at which the material yields and the contact response shift from elastic to plastic. The elasto-plastic regime of the model assumes a truncated pressure distribution in which the pressure is limited at the yield limit resembling an plastic deformation [27]. Lippke et al. found that combining the Thornton-Ning elasto-plastic model with the bond model is well in agreement for the densification simulation of Li-ion battery anodes [28].

This study focuses on the development of a normal force model to simulate the force response of halide solid electrolyte ($\text{Li}_3\text{YBrCl}_5$) and nickel rich cathode active material (CAM), NMC-811 ($\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$) during densification. The NMC active material particles as well as the halide solid electrolyte particles are elasto-plastic in nature [3,29]. The NMC active materials behave elastically under compaction levels up to 0.10% of their radius and beyond the yield point, the contribution of plasticity must be considered [29]. Adhesive contact models like EEPA are neglected to consider the influence of bond model in the particulate structure. Bond model allows for the evolution of contact stiffness and bond strength as a function of coordination number and confinement which enables realistic stress redistribution [30]. Moreover, in the wet processed solid-state separator or cathodes, the particles are held together by binder additives. Neglecting the forces arising from these additives leads to an underestimation of electrodes mechanical integrity and structure. The bond model approach places a spring bond between particles that can transmit forces as well as moments. The bonds are broken when the stresses acting on the bonds are above the critical strength [31]. This also allows for the modeling of the additive binder network in simulating particle structure.

Due to the factors of high deformability of the halide solid electrolyte particles and densification processes involving higher pressures, it is important to consider the large deformation behavior in the DEM simulations [14]. Therefore, calculating the force response of halide solid-electrolyte particles in the separator and its interaction with the CAM particles in the cathode can be modelled using an elasto-plastic model along with the stress based multi-contact model to effectively capture the force response at high densification pressures.

This article is organized as follows: at first, a general overview of the discrete element method and the existing elasto-plastic model is presented [27]. Based on this, the formulation of the new contact model is defined in section 2. Furthermore, benchmarking of the defined model with 3-particle and 5-particle systems under uniaxial loading and unloading conditions are done to demonstrate the performance of the multi-contact elasto-plastic model in comparison to other available models. Additionally, the model is applied to the uniaxial densification simulations of an ASSB separator and cathode and compared with other existing contact models. Furthermore, the influence of compaction pressures on different parameters such as final film thickness, degree of densification, porosity etc. are studied. The simulated

structure is also studied for the influence of compaction pressure on the ionic conductivity of the separator film. We conclude our work in the last section and finish with some outlooks.

2. Methods and materials

2.1. Discrete element method

Discrete element method can be used to study the microscopic properties of the macroscopic particulate materials as originally developed by Cundall [32]. The particle deformations are defined by overlaps between particles and is used to compute contact forces between two particles. The basic assumption is that the particle contacts are independent of one another resulting in a local resolution of forces [32,33]. It is a straight forward approach to solve the translational and rotational equations of motion of a many interacting particles system and are defined as

$$m_i a_i = \sum_j (F_{nij} + F_{tij}) + m_i g \text{ and } I_i \omega_i = \tau_{ij} \quad (1)$$

where m_i , a_i , I_i , ω_i are the mass, acceleration, moment of inertia and angular velocity of the particle i respectively and F_{nij} , F_{tij} , τ_{ij} are the normal force, tangential force and torque acting on particles i and j respectively. The acceleration due to gravity, g has the value 9.8 ms^{-2} [32,34]. The force calculation can follow various models to define the force–displacement law [35–37].

2.2. Thornton-Ning normal contact force

In this contact model, the behavior of a particle – particle contact is calculated considering the radii R_i , elastic moduli E_i , and poisson ratios ν_i . In the beginning of the contact development, the forces are calculated according to Hertz theory,

$$F_{el} = \frac{4}{3} E^* \sqrt{R^*} \delta^3 \quad (2)$$

where, δ is the overlap between the particles in contact, E^* is the effective elastic modulus calculated from the E_i , E_j and ν_i , ν_j and the effective radius, R^* calculated from r_i , r_j using the relations

$$\frac{1}{E^*} = \frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j}, \frac{1}{R^*} = \frac{1}{r_i} + \frac{1}{r_j}, \quad (3)$$

the plasticity effect on the contact initiates when the radius of the circular contact surface formed between the particles i and j , a exceeds the contact radius, a_y at the yield point. The radius of the contact surface can be calculated from the force acting on using

$$a = \sqrt[3]{\frac{3F_{el}R^*}{4E^*}} \quad (4)$$

The critical yield pressure p_y , at which the onset of plastic deformation occurs determines the value of a_y . The critical yield pressure is also related to the critical yield normal force f_y and the corresponding overlap δ_y by

$$f_y = \frac{1}{6} \left(\frac{R^*}{E^*} \right)^2 (\pi p_y)^3, \delta_y = \frac{1}{4} \frac{R^*}{E^*} (\pi p_y)^2 \quad (5)$$

After the yield overlap, the force–displacement relationship is described as

$$F_{el-pl} = f_y + \pi p_y R^* (\delta - \delta_y) \quad (6)$$

This indicates that after the yield point, the force–displacement curve exposes a linear relationship with the contact overlap. During unloading, the effect of plasticity is considered until the yield point is reached. In this phase of unloading, the effective plastic radius R_p^* is used to calculate the unloading force.

$$F_{\text{unloading}} = \frac{4}{3} E^* \sqrt{R_p^* (\delta - \delta_R)} \quad (7)$$

where δ_R is the residual plastic overlap [27]. The tangential force is modeled like in the literature [32] and is coupled to the normal force via Coulombs law.

$$F_{\text{tangential}} = -\mu F_n \frac{\dot{s}_t}{|s_t|} \quad (8)$$

where μ is the friction coefficient, F_n is the normal force component defined in eq (6) and eq (7), and $\dot{s}_t$ is the tangential component of the relative velocity vector.

2.3. Multi-contact elasto-plastic force model

When a particle is compressed, it does not only deform locally at the contact zone but also undergoes lateral expansion due to the Poisson effect. This lateral deformation causes the particle to expand into neighboring contacts, generating additional force components that are not captured by independent pairwise contact calculations. In dense packings with high coordination numbers, these laterally induced forces become significant as each contact influences the deformation state at adjacent contacts. Consequently, the force at a given contact is not solely determined by the overlap at that contact but is also affected by the cumulative deformation arising from all other contacts acting on the particle. Neglecting these coupled deformation effects leads to an underestimation of contact forces and inaccurate predictions of stress distribution within the system.

Consider the contact between the particles i and j . In classical DEM, the force response at the contact between particles i and j are calculated only considering the aspects of the contact of interest. However, in cases like compacting a pellet or calendaring a battery electrode, it is common that the particles experience high stresses and the method of force calculation by classical DEM is insufficient to effectively capture the force response at the contact of the particles. Therefore, all the subsequent contacts of the particles in considering contact should be considered for the calculation of the forces.

Here, a new contact model coupling the elasto-plastic model [27] and the stress-based multi contact model [24] is proposed to efficiently model the forces at the particle contact. The model assumes perfectly elastic particle behavior up to the yield overlap. Upon unloading within this regime, the particles respond elastically and exhibit no residual plastic deformation. Beyond the yield overlap, plastic deformation initiates, and the force response deviates from the ideal elastic force–displacement relationship.

A stress-based approach is used to calculate the effect of multiple contacts on a particle. As shown in fig. 2, the model utilizes the dyadic product of the length of the branch vector, l^n and the force at the contact, f^n and the sum of all such products from different contacts over the volume of the particle, V^p to calculate the stress tensor of rank 2.

$$\sigma^p = \frac{1}{V^p} \sum_{n=1}^{N^p} l^n \otimes f^n \quad (9)$$

The stress tensor accounts for the contributions of multiple contacts acting on a particle, thereby incorporating information from all interacting neighboring contacts. A multi-contact pressure term, obtained from the trace of this stress tensor is then used to compute the normal contact force within the DEM [24]

$$P_{ij} = (\text{tr}(\sigma_i) + \text{tr}(\sigma_j))/3 \quad (10)$$

In this concept, the new contact forces on each pair depends also on the trace of the stress tensor of neighboring particles, hence, an anisotropic deformation of a particle can be accounted for. Fig. 2

depicts a schematic concept of the approach. During the loading and unloading phase, the additional term aroused considering the pressure contribution from the simultaneous contacts is added to the normal force response of the elasto-plastic particles.

$$F_{mc} = (\beta \nu A_{ij}) P_{ij} \quad (11)$$

Introducing eq (11) into eq (2), eq (6) and eq (7),

$$F_{i \rightarrow j} = \begin{cases} F_{el} + F_{mc}, & \delta < \delta_y \\ F_{el-pl} + F_{mc}, & \delta \geq \delta_y \end{cases} \quad (12)$$

$$F_{i \rightarrow j} = F_{\text{unloading}} + F_{mc}, \delta \geq \delta_y \quad (13)$$

Eq (12) and eq (13) represent the normal force–displacement relationship of the multi-contact elasto-plastic contact model.

The developed contact model was implemented, and the simulations were performed using Ansys Rocky DEM.

2.4. Bond model

In order to define the electrochemically inactive materials in the system, bond model explained by Sangrós et al. is used [38]. In this model, a bond is considered as a cylindrical spring-dashpot connections between 2 particles i and j with radii r_i and r_j , when the distance between them, d_{ij} satisfy the distance criteria

$$d_{ij} < (r_i + r_j)(1 + f) \quad (14)$$

where, f is the control parameter that regulates the number of bonds formed in the simulation. The bond radius defines the cross-sectional area of the cylindrical connection through which the adhesive forces are transmitted. They are calculated according to the radii of the particles involved in bond formation, specifically the minimum of the two radii is considered. The radius of the bond r_b between 2 particles is defined by

$$r_b = \infty_b \min(r_i, r_j) \quad (15)$$

in which ∞_b is the parameter used to define the volume proportion of the bonds corresponding to the volume proportion of the inactive material used in the experiment, i.e., binder system for the solid-state separator and carbon black and binder in the electrode structure. At the time of activation, the bonds are in undeformed state and afterwards, any relative motion of the bonded particles will deform the bonds and to which reaction forces and moments are exerted to oppose the movement. As explained by Sangrós et al., the bonds normal and tangential stiffness S_n and S_t respectively, determines the forces $F_{b,n}$, $F_{b,t}$ and moments $M_{b,n}$, $M_{b,t}$ in normal and tangential directions respectively

$$\frac{dF_{b,n}}{dt} = -\nu_n S_n A \quad (16)$$

$$\frac{dF_{b,t}}{dt} = -\nu_t S_t A \quad (17)$$

$$\frac{dM_{b,n}}{dt} = -w_n S_n \frac{J}{2} \quad (18)$$

$$\frac{dM_{b,t}}{dt} = -w_t S_t J \quad (19)$$

where ν_n and ν_t are the relative velocities and w_n and w_t are the relative angular velocities between the bonded particles in normal and tangential directions respectively. J denotes the polar moment of inertia and A the cross-sectional area of the bonds.

The detailed definition of the bond model used in this study is explained in the studies published by Sangrós et al. [38] and Lippke et al. [28].

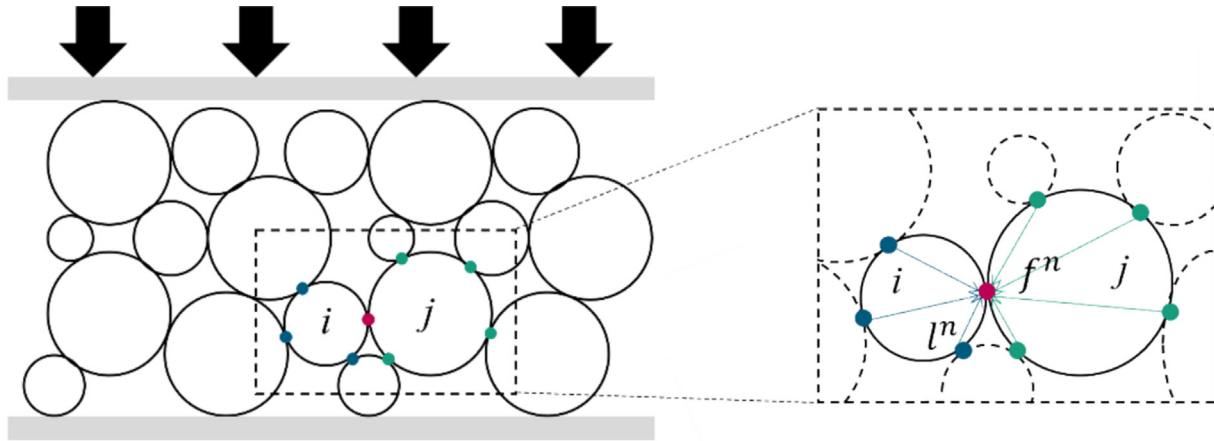


Fig. 2. Schematic of particle packing and details of the multiple contacts consideration by the model while calculating the force response at a contact. Snippet explaining the force and branch vectors for calculating the particles' stress tensor.

2.5. Modeling the ionic conductivity of solid-state separator

Within an ASSB, the transport of Li^+ ions rely on the network of solid electrolyte particles to reach the cathode side. To assess the electrochemical properties of the simulated structure of the separator, a particle contact network based on the densified particulate structure of the solid-state separator is determined. This network uses resistors to represent the particles' intrinsic resistance to ion transport.

All particle contacts are considered to accurately model the electrochemical properties of the microstructure. The contributions from the bonds are also considered for such calculations. In the case of a separator, the bonds generally represent the inactive binder content in the structure, and it does not contribute into the electrochemical properties of the structure. The developed post-processing script based on the work of Sangrós et al. [39] is used. The path length between the contacts inside a particle is considered to efficiently choose the most desirable path of conduction and an area dependent conductivity term is added to the system according to the depth of overlap of each contact. Here, in Fig. 3, a 4-particle system is defined in which different possible conduction paths are defined. The intrinsic resistance, R_p , of the particles is defined as a length dependent resistance term between the contacts of a particle. The resistance at the contacts, R_c , is dependent on the area of contact. The bond resistance, R_b , is considered for the representation of the binder's resistance to the ionic conduction. These intraparticle intrinsic ionic resistance and interparticle contact resistance forms a network of conducting paths for the ion

transport. The appropriate conducting path is the network between particles that links the high and low potential plates which confines the particulate structure and, through the linear system solution, automatically yields the path of lowest overall electrical resistance consistent with the full resistor network [39].

2.6. Materials

The solid-state separator slurry consists of halide SE (Saint-Gobain) and styrene-butadiene rubber (SBR) (Buna SL 4525; Arlanxeo Holding B.V) as binder to ensure structural integrity and mechanical stability. While the cathode slurry consists of NMC811(Umicore) as CAM, halide SE, CB (C-ENERGY SUPER C65; Nanografi) and SBR. Toluene (Sigma-Aldrich) is used as the solvent. Different properties of the materials used in the simulation are defined in the Table 1.

2.7. Experimental methods

2.7.1. Separator and cathode production

To validate the numerical method, ASSB separators and cathodes were manufactured with different densification pressures. In case of the cathode, the solid components are dry-mixed in a tumbling mixer (TURBULA® T2F; Willy A. Bachofen GmbH). SBR is dissolved in Toluene using a magnetic stirrer (Phoenix Instrument GmbH) to prepare the binder solution. The solid content amounts to 60 wt-% of the slurry. The slurry is dispersed in the binder solution using a Dispermat® (VMA-GETZMANN GmbH). The

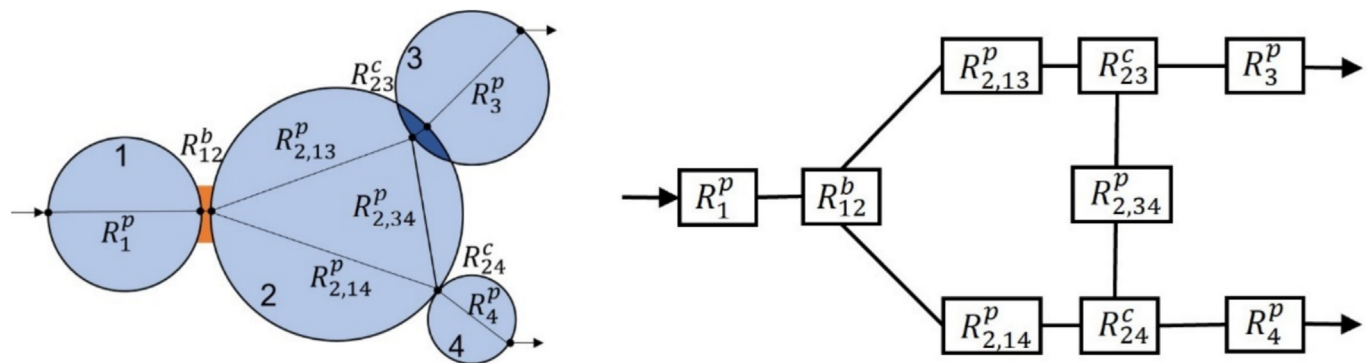


Fig. 3. Illustration of the arrangement of a connected particle structure and its analogous resistance network.

homogenized slurry is then coated on the substrate using a film applicator (COATMASTER 510; ERICHSEN GmbH & Co. KG) utilizing the doctor-blade technique and dried under Argon atmosphere at room temperature.

2.7.2. Densification

Circular discs with a diameter of 16 mm are precisely punched from the dried films of separator and cathode are subsequently positioned within a specialized pressing tool setup ensuring the separator specimen remains securely in place as illustrated in Fig. 4 (left). Uniaxial pressure is applied to the films using a hydraulic press (FKW GmbH), with the applied pressure systematically increased from 100 MPa to 350 MPa to study its effects. The thickness and mass of the separator and cathode film punchouts are meticulously measured both prior to and following the compaction process. The porosity before and after densification is calculated with the measured parameters using

$$\varepsilon = 1 - \frac{m}{\rho_{\text{eff}} t A} \quad (20)$$

where, ε is the porosity, m is the mass, ρ_{eff} is the effective density, t is the thickness and A is the surface area of the punchouts.

Initially, the film exhibits a thickness denoted as h_a , which is reduced to h_b under the compressed state. Due to the elastic recovery behavior of the binder matrix, the films undergo slight relaxation post-compaction, resulting in a final thickness of h_c . This relaxation phenomenon highlights the viscoelastic properties of the SBR binder and its influence on the structural characteristics of the films.

2.7.3. Ionic conductivity measurements

The punched specimens, each with a diameter of 16 mm, are placed within an insulating PTFE chamber. Two stainless-steel pistons serve as ion-blocking electrodes. The pistons are sealed using O-rings to ensure protection against air and humidity. The PTFE chamber is mounted within a cell-press device, which allows the application of pressure to the stainless-steel pistons and, consequently, to the separator specimen. During the measurements, a pressure of 25 MPa is applied using the cell-press device. Electrochemical impedance spectroscopy (EIS) measurements are performed using the BioLogic VMP300. The frequency of the applied signal varies over a wide range, typically from high (7 MHz) to low (1 Hz), to probe different electrochemical processes occurring within the system. The impedance, which is the ratio of the applied

voltage to the measured current, is recorded as a function of frequency. This allows for the identification and quantification of various resistive and capacitive elements in the system. The Nyquist plots obtained from the PEIS measurements are analyzed using an equivalent circuit modeling approach. The equivalent circuit is designed to represent the electrochemical behavior of the system, and the fitting process yields parameters such as resistance and capacitance. Specifically, the resistance corresponding to the ionic resistance of the separator specimen is extracted. This resistance value is then used to calculate the ionic conductivity of the separator using the following equation

$$\sigma = l / (R a) \quad (21)$$

where σ is the ionic conductivity, l is the thickness of the separator, R is the ionic resistance obtained from the equivalent circuit fitting, and a is the cross-sectional area of the separator.

2.8. Numerical methods

2.8.1. Representative volume element (RVE)

The experimental observations for the thickness of the films are considered on a $100 \mu\text{m} \times 100 \mu\text{m}$ surface area to define the representative volume element (RVE) of the separator and cathode films which is used as the simulation domain and are depicted in Fig. 4 (right). The light grey particles represent the halide solid electrolyte particles, and the dark grey particles represent the NMC particles. The binder matrix is defined using the bond model. The parameters of the bond model are calibrated according to the force response and volumetric proportion of the binder matrix in the RVE. The sizes and the mass of the inserted spheres are based on the particle size distribution (PSD) of the particles described in Table 1 and Table 2 respectively. The mass loading of the simulation microstructure is according to the mass loading of the experimental counterpart, and it corresponds to 0.03 g/cm^2 and 0.037 g/cm^2 for the separator and cathode respectively. This results in 36,442 and 72,964 particles in the simulation domains of the separator and cathode respectively.

2.8.2. Densification

In the densification simulation, the microstructure of the separator and the cathode is defined according to the initial thickness and porosity of the separator and the cathode films as obtained from the experiments before compaction. The particles are placed between two plates, one on the top of the particles resembling the

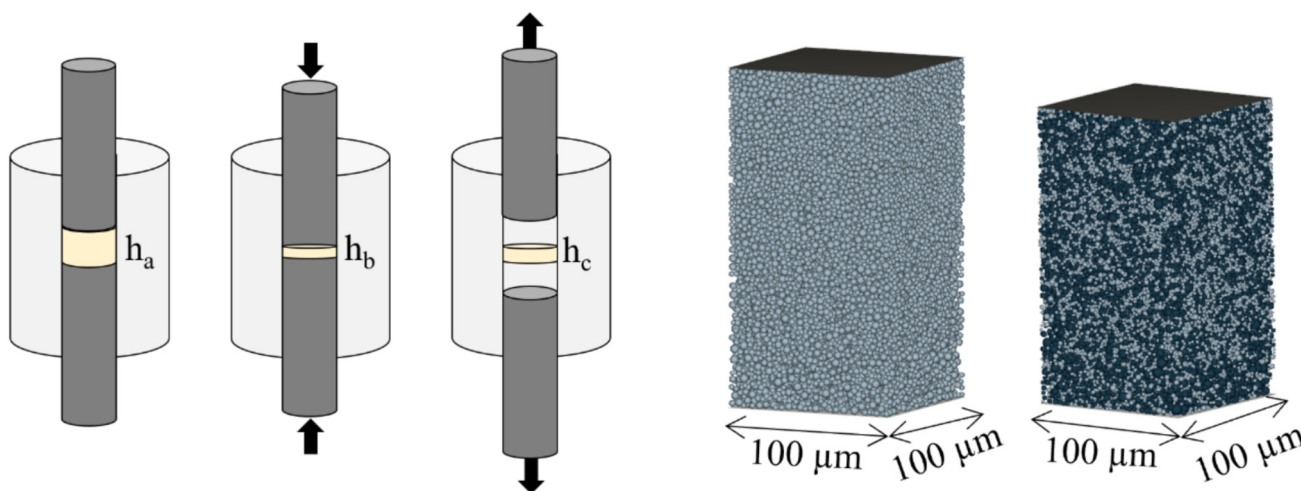


Fig. 4. (left) Schematic representation of the experiment setup used for the densification study, (right) RVE of the separator and cathode films used in the simulation setup.

Table 1
Modeling parameters of the densification simulation.

	Halide solid electrolyte	CAM	Unit
Particle size distribution	$d_{90} = 3.8$ $d_{50} = 2.1$ $d_{10} = 1.1$	$d_{90} = 6.1$ $d_{50} = 3.4$ $d_{10} = 2.6$	μm
Youngs modulus	10,58	140	GPa
Poisson ratio	0.3	0.25	–
Density	2.6	4.75	kg m^{-3}

Table 2
Solid-state separator and cathode formulation.

Material	Mass fraction	
	Separator	Cathode
Halide SE	97 wt-%	17.43 wt-%
SBR	3 wt-%	3 wt-%
CAM	–	77.6 wt-%
CB	–	0.97 wt-%

moving plate of the hydraulic press that compacts the separator and the cathode films and one on the bottom. In the case of cathode, the bottom plate is an Aluminum current collector and for the separator is the static plate of the hydraulic press. The top plate is moved down with a strain rate of 0.283 s^{-1} allowing to push the particles and compress them together until desired compression pressure is reached. At this point, the least thickness is reached. After this, the plate is moved further away from the particle confinement allowing for any relaxations of the particle bed. The final thickness of the particle bed is measured to compare with the experimental setup.

2.8.3. Binder content in the simulation

Lippke et al. reported that there is an increase in the binder volume fraction towards the electrode surface [40]. The gradient gets stronger as the areal loading is increased. Therefore, in the simulations the ∞_b is adjusted in such a way that the volume fraction of the binder reaches the target values of the inactive material in the experiments. In the experiments, the binder volume in a specimen is estimated according to the weight of the specimen and its corresponding wt.-% in the slurry used for coating the films. For the cathode simulations, a mixture of binder and carbon black is used and is represented by bonds. The total volume of these two components in an experimental specimen is used as a target value for the bond volume in the simulation. The simulation domain is divided into two equal parts and the value of ∞_b is regulated in each of it to achieve the same volumetric presence of the binder as in experiments and is explained in Table 3.

2.8.4. Simulation setup for ionic conductivity measurements

The simulated densified particle structure of the separator film is imported into a new simulation file with plates on top and bottom of the domain. Here, the plates represent the stainless-steel pistons as in the experimental setup. The bottom plate is kept stationary, and the densified particle bed is imported directly onto it. The top plate is slightly away from the particle bed and is moved slowly towards the particle bed until the experimentally used stacking pressure of 25 MPa is achieved. The resulting particle

Table 3
Values of the parameter ∞_b used in the simulations.

	Separator	Cathode
$\infty_{b, \text{bottom}}$	0.45	0.35
$\infty_{b, \text{top}}$	0.2	0.25

structure is used for the definition of particle network and contact detection. This network is then used to calculate the ionic conductivity of the separator film using the previously discussed method.

3. Results and discussions

3.1. Benchmarking of the force model using simplified particle systems

In particulate separators or cathodes, numerous interparticle contacts are formed between neighboring particles, and their number is substantially increased during densification. As densification progresses, particles are driven into closer proximity, leading to changes in both contact area and coordination number within the microstructure. To illustrate this effect, systems of 3 and 5 particles, corresponding to coordination numbers of 2 and 4, respectively, are investigated to demonstrate how the contact forces increase with increasing number of contacts. Consequently, it is essential to investigate the influence of different contact models in capturing the realistic force response at particle contacts during densification.

In this section, a test case of 3-particles system is used to study the influence of different contact models using a comparison of the force–displacement response. In the 3-particle system, a central particle is surrounded by 2 other particles on the top and bottom and is loaded and unloaded uniaxially and is shown in Fig. 5 (left). A contact on the central particle is chosen for the analysis. In Fig. 6 (left), it is evident that the Hertz model follows the same path during the loading and the unloading regime due to its elastic nature of calculations and over-estimates the force response at the contact. In Thornton–Ning model, the force–displacement curve deviates from the Hertz curve after the yield overlap is reached. However, at higher compaction pressures, where multiple particles are simultaneously in contact, the Thornton–Ning contact model within the classical DEM framework underestimates the particles' force response, as it neglects the influence of multiple contacts acting on a single particle. In Fig. 6 (left), it is observed that the additional term including the pressure term evolved from the multiple contacts influence the calculation of force as well as residual overlap in the system. The multi-contact elasto-plastic model is efficient in calculating realistic force response of the particle at high stresses because the pressure term is highly influenced by the stresses on the particles. In the lower ranges of the stresses, the additional pressure terms might not be contributing much into the loading force calculations. Thus, for compaction simulations with higher pressures, multi-contact elasto-plastic model can precisely calculate the force response at particle contacts during the densification of cathode and separator particulate structures.

To study the influence of number of contacts on a particle during the force calculation, a system with 5 particles is utilized in which a similar setup as above with particles on top, bottom, left and right of a central particle is used, as shown in Fig. 5 (right). In this case, additional contacts are present on the central particle apart from the contact that is already considered for the force and overlap calculation. The stresses on the particle due to the influence of the additional contact increases the amount of force calculated. In Fig. 6 (right), the force response at a particle contact in a 3- and 5-particle system is compared and it is visible that the system in which the particle has more contacts have a significant difference in the calculated force. This explains the importance of deviating from classical DEM and considering multiple contacts on a particle during the force calculation. It is understood that higher the number of simultaneous contacts on a particle, higher is the influence of the additional pressure term in the force calculation and hence the residual plastic overlap in the system. Here, the overlap (%) is relative to the radius of the particle and is calculated as in (22),

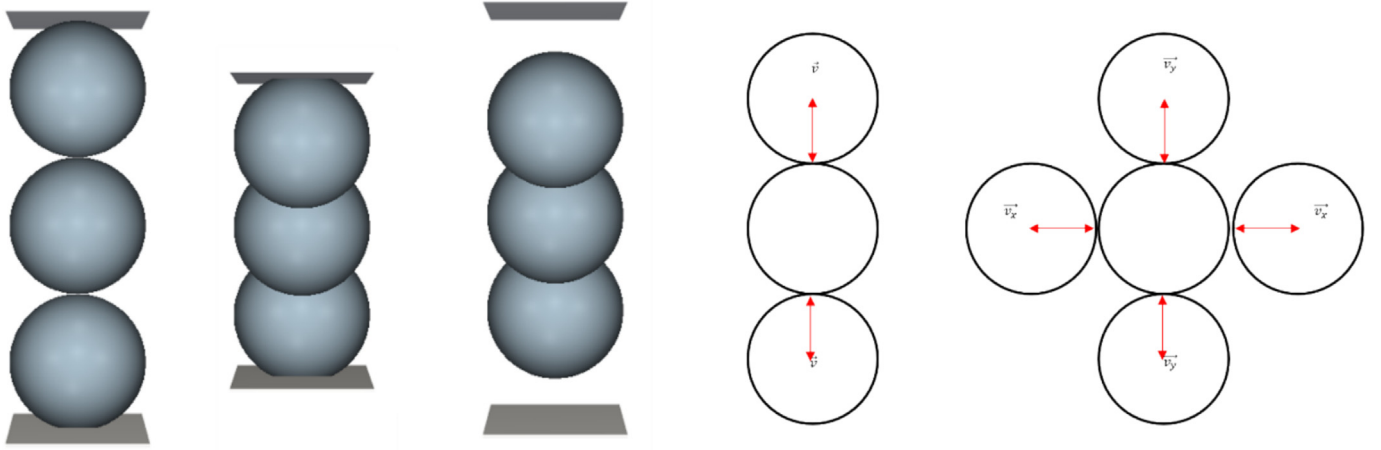


Fig. 5. (left) A system with 3 particles used for studying the force response of the model, the centre particle being loaded and unloaded by the particles on top and bottom, (right) systems with different number of particles to study the influence of contacts in force calculation.

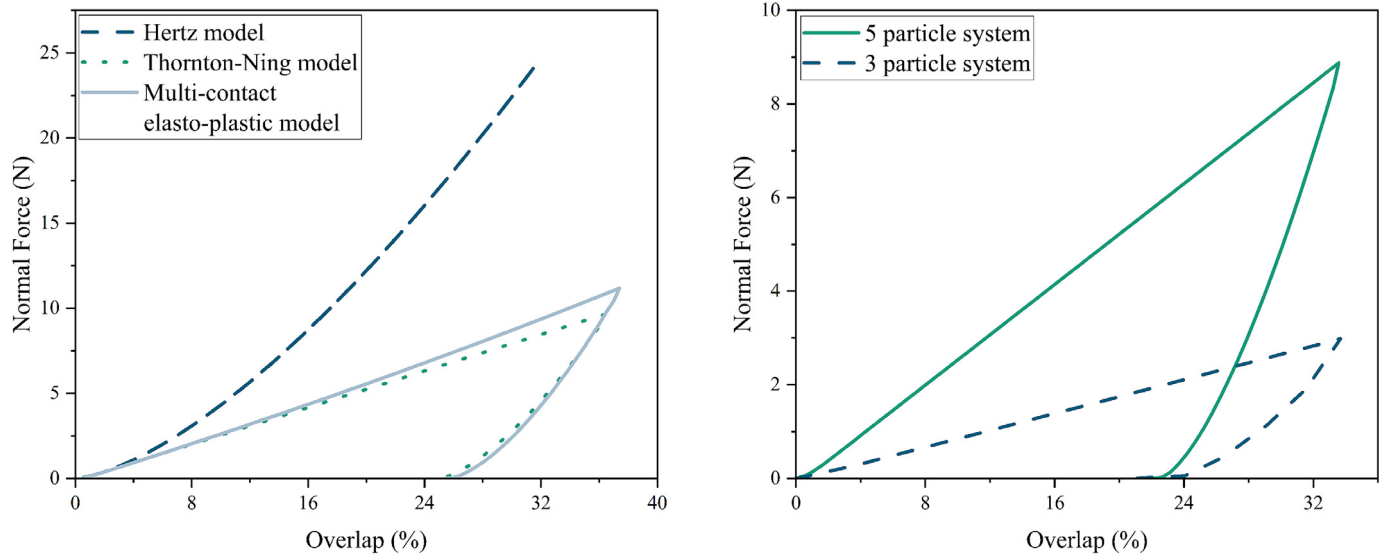


Fig. 6. (left) Influence of different contact models in the force–displacement relationship of a contact. (right) Influence of number of contacts in the force–displacement relationship using multi-contact elasto-plastic model.

$$\delta(\%) = \frac{\delta}{\min(r_i, r_j)} \times 100 \quad (22)$$

3.2. Densification

3.2.1. Solid-state separator

Halide solid electrolyte particles with the defined unimodal PSD in Table 1 is used in the separator densification simulations along with the SBR binder. Halide particles are represented with spherical shape while the SBR binder is represented by the mass less bonds in the simulation. However, the volume fraction of the SBR binder in the simulation structure is regulated using the parameters in the bond model. Within the bond model, dimensionless bond damping coefficient is fitted 0.9997 and the normal, S_n and tangential bond stiffness, S_t corresponds to $2.5e^{+13}\text{N/m}^3$ and $1.875e^{+13}\text{N/m}^3$ respectively are empirically chosen. The yield ratio, of the multi-contact elasto-plastic model is set to 0.0103 for the interaction between particles and the geometrical empirical factor

for the multi-contact pressure term, β is calibrated to 0.5. The choice of this factor is made with compromise that, otherwise the time step has to be further decreased for an efficient calculation and thereby increasing the computational expense.

The insertion of spherical particles is done volumetrically in the domain according to the areal loading of the separator films as observed from the experiments. The particles are placed randomly in the domain to satisfy the mass and porosity of the experimentally characterized twin. In the virtual separator setup, the bond model is activated in the very first timestep to ensure the effect of SBR binder during the densification and relaxation of the film. The simulations were carried out with a control over applied stress, i.e., the upper plate is pressing the particle bed until the experimentally observed pressure is reached.

The setup is calibrated using a reference case in which the particle structure is compressed to an applied pressure of 350 MPa. Subsequently, the plate is retracted, allowing the particle bed to relax and establish the final membrane thickness. The densification simulation scheme is illustrated in Fig. 7.

To study the influence of different contact models on the compaction simulations, the separator particle bed is subjected to simulations with Hertz model [36], Thornton-Ning model [27] and multi-contact elasto-plastic model. As shown in Fig. 8, the simulated film thickness using different models are compared with the experimental thickness. In all cases, the simulation was done for a fixed amount of time by moving the upper plate with a constant velocity, such that the particles are compacted together until the defined time is reached before the plate is moved away. In the case of Hertz model, the final film thickness calculated by the simulation is way higher than the experimentally observed value. Also, the compaction pressure calculated by the Hertz model to reach this final thickness is significantly higher than that applied in the experiment. Employing the Thornton-Ning model in the same simulation setup estimates the applied pressure at 300 MPa, resulting in a film thickness deviating from the experimental observation. The curve deviates from the Hertz curve once the yield point is reached and behaves more plastic exhibiting a much lower applied pressure than the results from Hertz model. For the case using multi-contact elasto-plastic model, the effect of the multi-contact term arises as the applied pressure is higher and thus increasing the number of contacts on the particles. For relative densities of 0.7 and higher, contact dependency becomes significant, and multiple-contact DEM models are necessary [41]. The final film thickness given by the simulation was well comparable with the experimental values while precisely calculating the normal compaction pressure to 350 MPa, which is also experimentally used. Thus, using Hertz model and Thornton-Ning model for compaction simulations of systems with high relative densities are not recommended while the multi-contact elasto-plastic accurately predicted the film thickness for the exact same normal compaction pressure.

Based on the insights gained from analyzing the influence of different contact formulations on the simulated densification response, the subsequent investigations employ the previously introduced multi-contact elasto-plastic model. A series of DEM simulations were performed on the same RVE of the separator film, in which a constant displacement rate of 0.00707 m s^{-1} on the pressing plate was applied while the loading duration systematically varied to reproduce the compaction pressures applied in the experiments. This also allows for a quasi-static event and thus

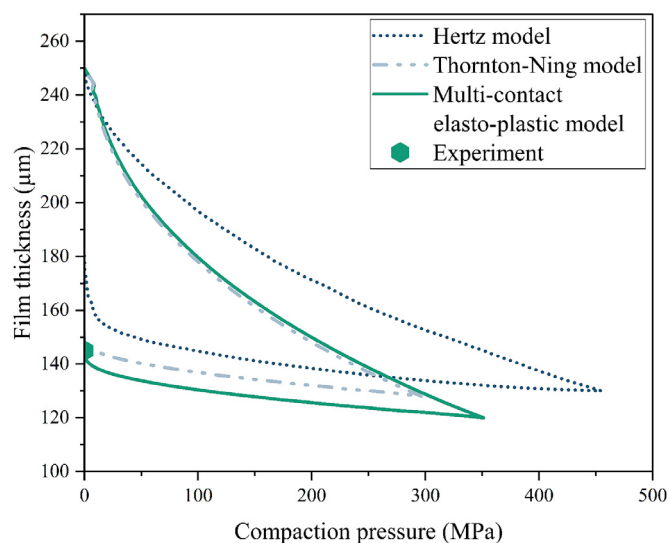


Fig. 8. Influence of different contact models on the compaction pressure on the particle structure and the final film thickness.

negate any inertial forces. Target maximum pressures of 100 MPa, 200 MPa, 300 MPa, and 350 MPa were imposed, and the corresponding compressed configurations were extracted once the respective pressure levels were attained. After reaching the prescribed pressure, the pressing plate was retracted to allow for structural relaxation of the particle assembly. The final film thickness was subsequently evaluated and is presented in Fig. 9.

The degree of densification (DoD) quantifies the extent to which the particle bed compacts under an applied pressure. It is defined as the relative reduction in film thickness, expressed as the difference between the initial and final thickness normalized by the initial thickness. As shown in Fig. 10, both porosity and DoD curves exhibit a distinct change in slope for compaction pressures lower and higher than 100 MPa, reflecting the transition from elastic to plastic deformation that resembles the change in the stiffness. Notably, the experimentally measured densification behavior is in close agreement with the predictions obtained from the DEM simulations. This consistency further validates the capability of

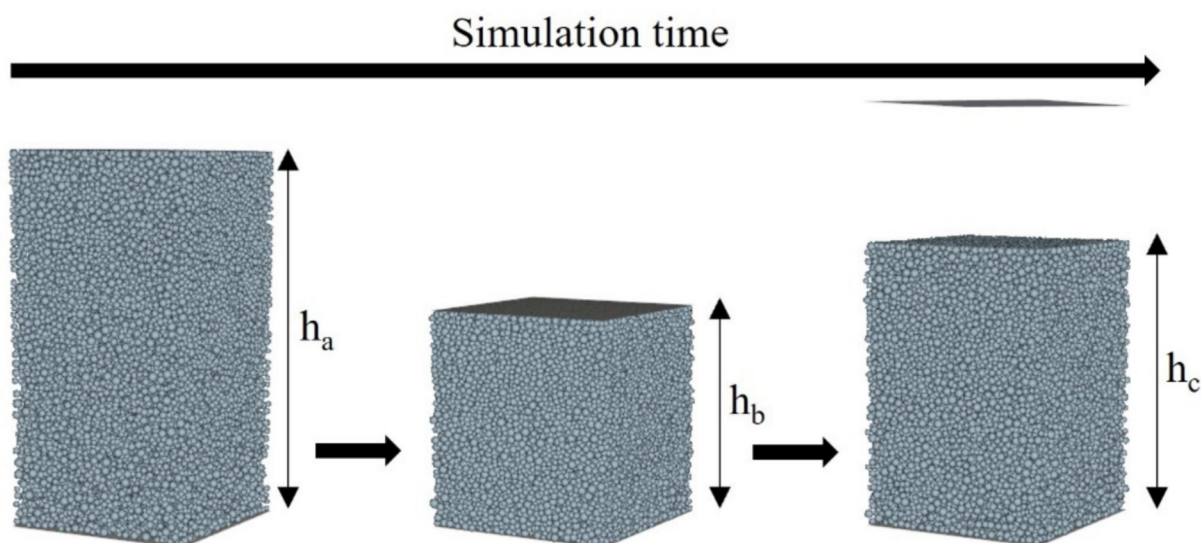


Fig. 7. Visualization of the simulated densification process of the solid-state separator. h_a is the initial thickness of the coating, h_b denotes the compacted thickness at maximum applied pressure, and h_c is the final film thickness after the electrode relaxes from the pressure applied by the press.

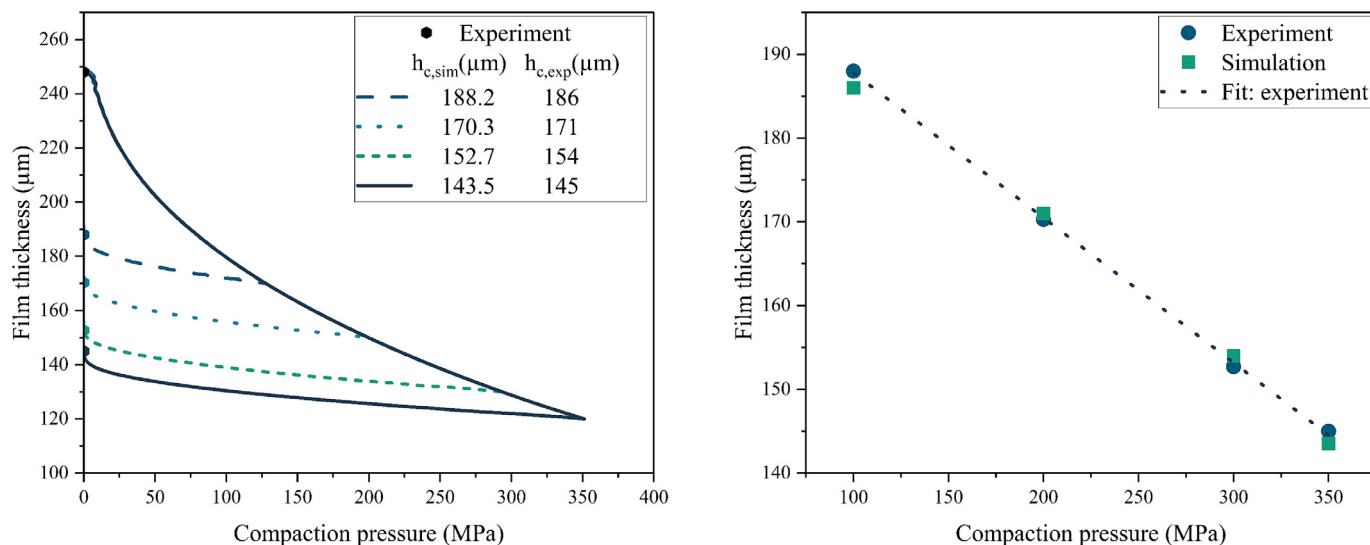


Fig. 9. (left) Film thickness of the solid-state separator over compaction pressure obtained by densification simulations, (right) comparison of final film thickness obtained from experiments and simulations.

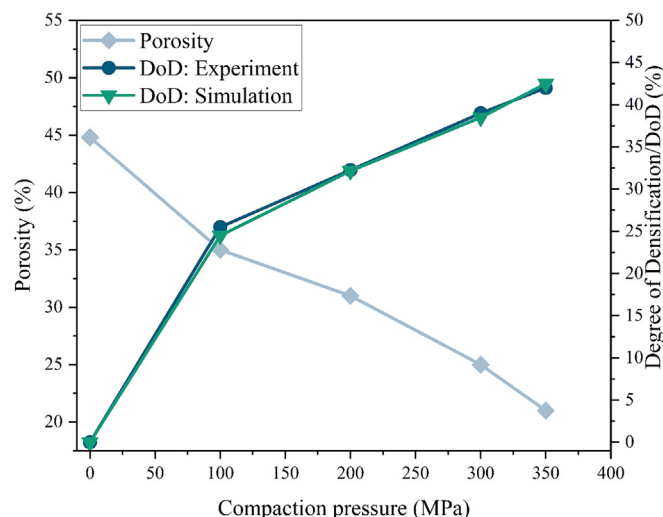


Fig. 10. Influence of different densification pressure on the porosity and evolution of the degree of densification of the final structure of the solid-state separator.

the proposed stress-based multi-contact elastoplastic model to accurately capture the mechanical response of halide-based solid electrolyte films subjected to high compaction pressures.

3.2.2. Cathode

The cathode in an all-solid-state battery is made of a composite of cathode active material, solid electrolyte and a conductive binder. For the cathode simulations, the CAM and solid electrolyte particles are considered as spherical particles and the conductive binder is defined using the mass less cylindrical bonds between the active materials and the solid electrolyte particles. The cylindrical bonds also represent the conductive binder matrix in the cathode volumetrically. That means, the volume fraction of the SBR binder and the CB calculated experimentally is considered while choosing α_b in the bond model. In the simulations similar considerations are made to insert the CAM and electrolyte particles. The corresponding volume fractions of the conductive carbon binder mixture is illustrated by the bond volume using the bond model. Fig. 11 depicts the simulation scheme of a cathode film

under uniaxial pressure. In the case of cathode, the bottom plate in the simulation structure is the Aluminum current collector to which bonds are established to ensure that the cathode film stays adhered to the current collector. The simulation setup is similar to that of the separator film as explained above.

A series of DEM simulations was conducted on an identical representative volume element (RVE) of the composite cathode film, in which the displacement rate of the pressing plate was systematically adjusted to reproduce the compaction pressures applied in the experiments. Target maximum pressures of 125 MPa, 200 MPa, 300 MPa, and 350 MPa were imposed, and the corresponding compressed states were recorded once the respective pressure levels were reached. Subsequently, the pressing plate was retracted to allow for elastic-plastic relaxation of the particle assembly. The resulting final film thickness was then determined and is reported in Fig. 12. Here, the solid electrolyte particles which are small compared to the CAM rearrange its position due to the particle motion arising from the compaction process. The rearrangement is such a way that the electrolyte particles fill in the void space between the CAM particles. This rearrangement happens in the early stages of pressure applications and thus a sudden change in the thickness of the cathode films within low pressure regions. This phenomenon is visible also in both simulations and experiments and is illustrated in Fig. 12.

The degree of densification (DoD) is as a quantitative measure of the compaction experienced by the particle assembly under an applied pressure. As illustrated in Fig. 13, at low applied pressures, the mechanical response is governed by elastic deformation of the particle bulk, which is reflected in the initial steep slope of the curves; with increasing compaction pressure, the onset of plastic deformation at interparticle contacts is evidenced by a progressive change in the slope of the porosity and DoD curves. Importantly, the experimentally observed densification behavior shows good agreement with the corresponding DEM simulation results. This agreement provides further validation of the proposed stress-based multi-contact elastoplastic model and demonstrates its capability to reliably capture the mechanical response and densification behavior of halide-based solid electrolyte films under high compaction pressures.

For the unimodal PSD, the proposed model demonstrates close agreement with experimental data across the full compaction pressure range investigated, accurately capturing the nonlinear densi-

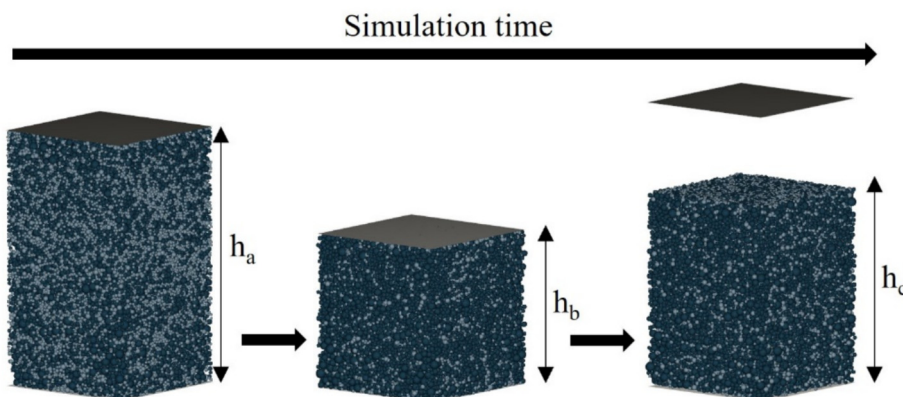


Fig. 11. Visualization of the simulated densification process of the cathode structure. h_a is the initial thickness of the coating, h_b denotes the compacted thickness at maximum applied pressure, and h_c is the final film thickness after the electrode relaxes from the pressure applied by the press.

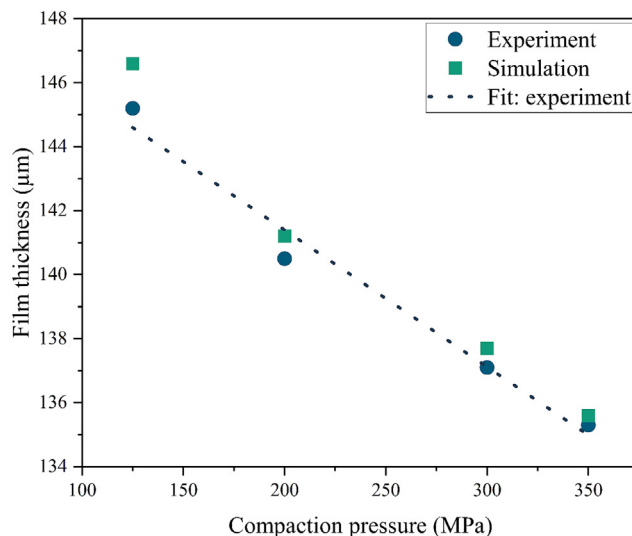
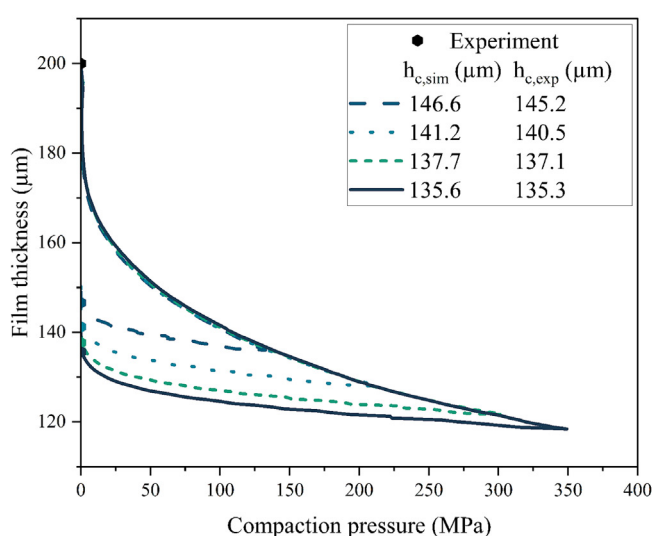


Fig. 12. (left) Film thickness of the composite cathode over compaction pressure obtained by densification simulations, (right) comparison of final film thickness obtained from experiments and simulations.

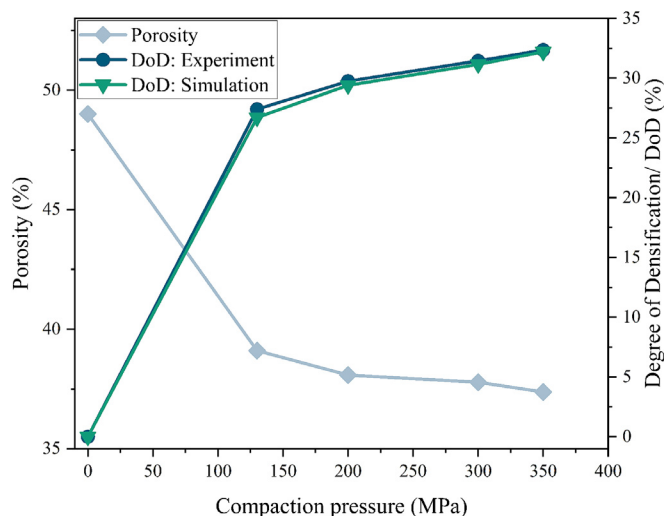


Fig. 13. Influence of different densification pressure on the porosity and evolution of degree of densification of the final structure of the composite cathode.

fication behavior characterized by the decreasing film thickness at higher pressures. However, for the bimodal PSD, the model captures the overall densification trend, though the more complex packing arrangement where the fine particles occupy the interstitial space between the larger particles creates high heterogeneity in local stress distribution.

3.3. Ionic conductivity measurements of solid-state separator

The densified separator structure with different compaction pressures from the simulations above are used as the particle structure for electrochemical characterization. The particle structure is confined between two plates acting as high and low potential geometries. A stacking pressure of 25 MPa is applied to the particle structure by moving the high potential plate towards the particle structure and is in accordance with the experimental procedure of ionic conductivity measurements of solid-state separator. The halide particles in the separator structure has an intrinsic ionic conductivity of 1.8 mS/cm [6,7]. The binder system represented by the bond model in the structure is electrochemically inactive and resistive to the ionic movement. Also, the intrinsic

sic resistance of the bonds is calibrated in accordance with the experiments. A post processing script as explained in Section 2.5 is employed to identify the particle contact network between the high and low potential geometries. The matrix with all the information extracted from the contact network is then solved according to the numerical procedure defined by Sangrós et al. [39].

Fig. 14 presents the numerically calculated values of the ionic conductivity of the separator films and is compared to the experimentally observed values of the ionic conductivity measurements, and the percentage of surface area in contact with respect to the densification pressure.

The percentage of surface area in contact between the particles is calculated according to the contact area defined by the model. It is a measure of particles contact area to the total area of the particle surface. It is understood that the number of contacts and the contact area between the particles increase as the densification pressure increases. Since the force calculation is dependent on the contact model, the contact area between the particles also varies depending on the contact model. With the increasing pressure, the number of contacts between the particles increases while reducing the porosity of the structure. This means that there are more ion conducting pathways available in the structure for easy ion transport. This is also evident from Fig. 14, as the densification pressure is increased, the ionic conductivity of the structure also increases. Specifically, in this case, the ionic conductivity of the separator made with Halide solid electrolytes increases noticeably in the structures densified with a compaction pressure of 200 MPa. This is observed both in the experimental and simulated structure. The difference in the experimental and simulated values of ionic conductivity for low densified structures are relatively high to that of the values for the structures densified with higher pressure. It is because of the increase in particles in contact and the increase in the contact area between the particles. Even though the exact values of ionic conductivity extracted from the simulation differs a bit from the experimental values, the trend in its values dependent on the densification pressure is similar to what is observed in the experiment. This difference in values can be due to the approximation of particle shapes used in the simulation. Even though the halide solid electrolyte particles are irregular in shape, for the simplification and the primary aim of the simulation to develop a new contact model to model multi-contact elasto-plastic behavior is satisfied, a compromise on the choice of particle shape is made. A new set of study employing more realistic particle shapes can be done in future.

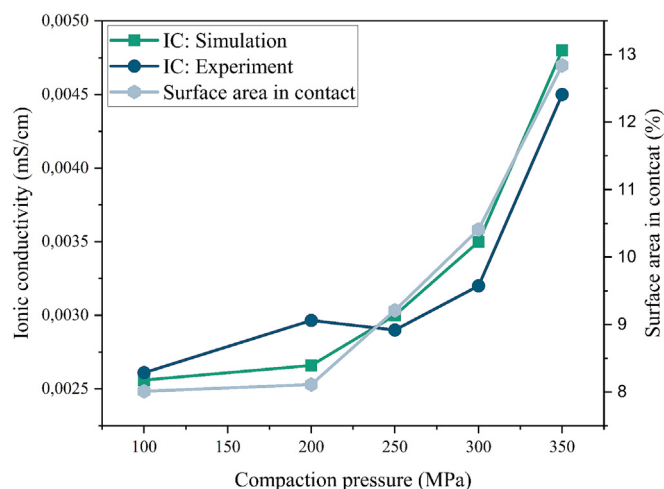


Fig. 14. Influence of densification pressure on the ionic conductivity and surface area in contact between solid electrolyte particles of the solid-state separator.

4. Conclusion

The advancement of high-performance ASSBs requires achieving high relative densities of separator and cathode to enable efficient ionic and electronic transport and minimize interfacial resistance. In this study, a model is presented to study the influence of multiple contacts on an elasto-plastic particle while determining the force–displacement relationship vis discrete element method (DEM). The model was first validated using controlled benchmark configurations designed to isolate multi-contact effects: (i) a three-particle compressed between rigid plates, representing a predominantly one-dimensional contact chain, and (ii) a multi-contact arrangement in which a central particle is subjected to concurrent loading from multiple neighbors, representative of dense contact networks at high packing fraction. These benchmarks demonstrated that multi-contact conditions produce (a) additional force contributions beyond independent pairwise interactions and (b) an influence on the residual overlap after unloading, confirming the necessity of incorporating contact coupling when plastic deformation is active. The framework was then applied to densification simulations of solid-state battery components, including a separator and a composite cathode, starting from an initial porosity of approximately 45%. The model is compared with the other contact models such as Hertz elastic normal force model and Thornton-Ning elasto-plastic model along with the bond model for the densification simulations of solid-state separator. The comparison showed that the proposed approach yields accurate predictions of the compaction response, including compaction pressure and final film thickness, with close agreement to experimental measurements. The conclusions made from the model comparison simulations were subsequently used for the densification simulation of the solid-state separator and the composite cathode of a halide-based system. The final film thickness and degree of densification were validated against experimental results. The proposed model successfully reproduces microstructures with a final porosity of 20%. While simulations at porosities below 20% are in principle feasible, they were not pursued in this study due to the significantly increased computational cost associated with the smaller time steps required to accurately capture the force response in high-density pressure regimes. The proposed model captures the stronger non-linearity at higher stress regions and improves agreement with experiments in the dense regime. Finally, the densified structure of the solid-state separator is used for electrochemical characterization and is validated with experiments, once again providing a good agreement to the calculated residual overlap by the proposed model. To conclude, the proposed model can realistically capture the plastic deformation of the particle structure and can be used for the densification simulations of battery components. Future work will focus on extending the model to simulate structures with porosities below 20%, incorporating mechanical responses during electrochemical cycling to establish an approach for predicting cell-level performance based on microstructural characteristics.

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CRediT authorship contribution statement

Cerun Alex Varkey: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project adminis-

tration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kostas Giannis:** Writing – review & editing, Methodology, Conceptualization. **Sebastian Melzig:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Carsten Schilde:** Supervision, Resources, Conceptualization. **Sabrina Zellmer:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Validation of Multi-contact elasto-plastic model using Finite Element Method

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References

- [1] Z. Zhang, Y. Shao, B. Lotsch, Y.-S. Hu, H. Li, J. Janek, L.F. Nazar, C.-W. Nan, J. Maier, M. Armand, L. Chen, New horizons for inorganic solid state ion conductors, *Energy Environ. Sci.* 11 (2018) 1945–1976.
- [2] J. Janek, W.G. Zeier, A solid future for battery development, *Nat. Energy* 1 (2016).
- [3] T. Asano, A. Sakai, S. Ouchi, M. Sakaida, A. Miyazaki, S. Hasegawa, Solid Halide Electrolytes with High Lithium-Ion Conductivity for Application in 4 V Class Bulk-Type All-Solid-State Batteries, *Advanced materials* (Deerfield Beach, Fla.) 30 (2018) e1803075.
- [4] S. Randau, D.A. Weber, O. Kötz, R. Koerver, P. Braun, A. Weber, E. Ivers-Tiffée, T. Adermann, J. Kulisch, W.G. Zeier, F.H. Richter, J. Janek, Benchmarking the performance of all-solid-state lithium batteries, *Nat. Energy* 5 (2020) 259–270.
- [5] A. Manthiram, X. Yu, S. Wang, Lithium battery chemistries enabled by solid-state electrolytes, *Nat. Rev. Mater.* 2 (2017).
- [6] P. López-Aranguren, P. Lannelongue, J. Carrasco, A. Golov, A. Robles-Fernandez, E. Gonzalo, A. Tron, T. Marchandier, A.-K. Hatz, R. Fathi, Q. Xu, D. Chaykina, C.A. Varkey, C. Gavagnin, M. Kaminski, A. Almousli, G.G. Eshetu, N. Hamzelui, F. Diaz, E. Drude, S. Abada, N. Guy, A. Sudharshan, M. Brunetti, S. Beschmitt, HELENA project: Driving innovation in high energy density Li-metal halide solid-state batteries for electric vehicles and aircrafts, *Comput. Struct. Biotechnol. J.* 29 (2025) 72–84.
- [7] P. Lannelongue, S. Lindberg, E. Gonzalo, A. Golov, F. Bonilla, J.M. Del Lopez Amo, T. Marchandier, A. Tron, J. Carrasco, P. López-Aranguren, Stable cycling of halide solid state electrolyte enabled by a dynamic layered solid electrolyte interphase between Li metal and Li₃YCl₄Br₂, *Energy Storage Mater.* 72 (2024) 103733.
- [8] J. Janek, W.G. Zeier, Challenges in speeding up solid-state battery development, *Nat. Energy* 8 (2023) 230–240.
- [9] P. Minnmann, L. Quillman, S. Burkhardt, F.H. Richter, J. Janek, Editors' Choice—Quantifying the Impact of Charge Transport Bottlenecks in Composite Cathodes of All-Solid-State Batteries, *J. Electrochem. Soc.* 168 (2021) 40537.
- [10] S.-Y. Lee, J.-H. Han, H.-W. Gong, J. Ahn, K.-W. Yi, Y.W. Cho, Characterization of densification behavior and mechanical properties of solid electrolyte powders for all solid-state batteries, *J. Mater. Chem. A* 13 (2025) 6342–6346.
- [11] R. Shang, T. Nelson, T.V. Nguyen, C. Nelson, H. Antony, B. Abaoag, M. Ozkan, C.S. Ozkan, A comprehensive review of solid-state lithium batteries: Fast Charging characteristics and in-operando diagnostics, *Nano Energy* 142 (2025) 111232.
- [12] P. Guichard, B. Mathieu, E. Woillez, Finite Element simulation of NMC Particle Fracture during Calendering: a Route to Optimize Electrode Microstructures, *Batteries Supercaps* 9 (2026).
- [13] N. Boaretto, I. Garbayo, S. Valiyaveetil-SobhanRaj, A. Quintela, C. Li, M. Casas-Cabanas, F. Aguesse, Lithium solid-state batteries: State-of-the-art and challenges for materials, interfaces and processing, *J. Power Sources* 502 (2021) 229919.
- [14] X. Han, Y. Xu, H. Li, Z. Wang, J. Yue, X. Yan, S. Zhang, J. Fu, Y. Xia, L. Zhou, S. Wei, X. Liu, X. Wang, C. Zhao, X. Li, S.-H. Bo, J. Wang, X. Sun, J. Liang, Mechanically robust halide electrolytes for high-performance all-solid-state batteries, *Nat. Commun.* 16 (2025) 9770.
- [15] Y.-H. Cho, J. Wolfenstine, E. Rangasamy, H. Kim, H. Choe, J. Sakamoto, Mechanical properties of the solid Li-ion conducting electrolyte: Li_{0.33}La_{0.57}TiO₃, *J. Mater. Sci.* 47 (2012) 5970–5977.
- [16] M. Papakyriakou, M. Lu, Y. Liu, Z. Liu, H. Chen, M.T. McDowell, S. Xia, Mechanical behavior of inorganic lithium-conducting solid electrolytes, *J. Power Sources* 516 (2021) 230672.
- [17] H. Liu, X. Cheng, Y. Chong, H. Yuan, J.-Q. Huang, Q. Zhang, Advanced electrode processing of lithium ion batteries: a review of powder technology in battery fabrication, *Particuology* 57 (2021) 56–71.
- [18] F. Gilibert, J.-N. Roux, A. Castellanos, Computer simulation of model cohesive powders: influence of assembling procedure and contact laws on low consolidation states (2007).
- [19] S. Torquato, F.H. Stillinger, Jammed hard-particle packings: from Kepler to Bernal and beyond, *Rev. Mod. Phys.* 82 (2010) 2633–2672.
- [20] M. van Hecke, Jamming of soft particles: geometry, mechanics, scaling and isotaticity, *Journal of physics, Condensed Matter : an Institute of Physics Journal* 22 (2010) 33101.
- [21] N. Brodu, J.A. Dijkstra, R.P. Behringer, Multiple-contact discrete-element model for simulating dense granular media, *Physical review, E, Statistical, Nonlinear, and Soft Matter Physics* 91 (2015) 32201.
- [22] J. Zheng, M.O. Ciantia, An Efficient Damage-Plasticity DEM Contact Model for Highly Porous Rocks, *Rock Mech. Rock Eng.* 58 (2025) 5733–5754.
- [23] N. Ghods, P. Poorsolhjoui, M. Gonzalez, S. Radl, Discrete element modeling of strongly deformed particles in dense shear flows, *Powder Technol.* 401 (2022) 117288.
- [24] K. Giannis, C. Schilde, J.H. Finke, A. Kwade, M.A. Celigueta, K. Taghizadeh, S. Luding, Stress based multi-contact model for discrete-element simulations, *Granul. Matter* 23 (2021).
- [25] M. So, G. Inoue, K. Park, K. Nunoshita, S. Ishikawa, Y. Tsuge, Contact model for DEM simulation of compaction and sintering of all-solid-state battery electrodes, *MethodsX* 9 (2022) 101857.
- [26] S.C. Thakur, J.P. Morrissey, J. Sun, J.F. Chen, J.Y. Ooi, Micromechanical analysis of cohesive granular materials using the discrete element method with an adhesive elasto-plastic contact model, *Granul. Matter* 16 (2014) 383–400.
- [27] C. Thornton, Z. Ning, A theoretical model for the stick/bounce behaviour of adhesive, elastic-plastic spheres, *Powder Technol.* 99 (1998) 154–162.
- [28] M. Lippke, C. Willuhn, T. Ohnimus, T. Heckmann, P. Scharfer, W. Schabel, C. Schilde, A. Kwade, Coupled numerical simulation of the drying and calendaring for Lithium-ion battery anodes with non-spherical particles, *Powder Technol.* 452 (2025) 120566.
- [29] C. Sangrós Giménez, B. Finke, C. Schilde, L. Froböse, A. Kwade, Numerical simulation of the behavior of lithium-ion battery electrodes during the calendaring process via the discrete element method, *Powder Technol.* 349 (2019) 1–11.
- [30] Y. Yamaguchi, S. Biswas, T. Hatano, L. Goehring, Failure processes of cemented granular materials, *Phys. Rev. E* 102 (2020) 52903.
- [31] D.O. Potyondy, P.A. Cundall, A bonded-particle model for rock, *Int. J. Rock Mech. Min. Sci.* 41 (2004) 1329–1364.
- [32] P.A. Cundall, O.D.L. Strack, A discrete numerical model for granular assemblies, *Géotechnique* 29 (1979) 47–65.
- [33] L.E. Silbert, D. Ertas, G.S. Grest, T.C. Halsey, D. Levine, S.J. Plimpton, Granular flow down an inclined plane: Bagnold scaling and rheology, *Physical review, E, Statistical, Nonlinear, and Soft Matter Physics* 64 (2001) 51302.
- [34] Y. Tsuji, T. Tanaka, T. Ishida, Lagrangian numerical simulation of plug flow of cohesionless particles in a horizontal pipe, *Powder Technol.* 71 (1992) 239–250.
- [35] M. Gonzalez, A.M. Cuitiño, A nonlocal contact formulation for confined granular systems, *J. Mech. Phys. Solids* 60 (2012) 333–350.
- [36] R.D. Mindlin, H. Deresiewicz, Elastic Spheres in contact under Varying Oblique Forces, *J. Appl. Mech.* 20 (1953) 327–344.
- [37] L. Vu-Quoc, X. Zhang, An elastoplastic contact force-displacement model in the normal direction: displacement-driven version, *Proc. R. Soc. Lond. A* 455 (1999) 4013–4044.
- [38] C. Sangrós Giménez, B. Finke, C. Nowak, C. Schilde, A. Kwade, Structural and mechanical characterization of lithium-ion battery electrodes via DEM simulations, *Adv. Powder Technol.* 29 (2018) 2312–2321.
- [39] C. Sangrós Giménez, A. Pistorio, C. Willuhn, C. Schilde, A. Kwade, Modelling the electrical conductivity of Lithium-ion battery electrodes via a bonded-particle approach, *Adv. Powder Technol.* 36 (2025) 104931.
- [40] M. Lippke, T. Ohnimus, F. Frankenberg, C. Schilde, A. Kwade, Drying and calendaring of Lithium Ion battery electrodes: a combined simulation approach, *Powder Technol.* 444 (2024) 119984.
- [41] K. Giannis, C. Schilde, J.H. Finke, A. Kwade, Modeling of High-Density Compaction of Pharmaceutical Tablets using Multi-Contact Discrete Element Method, *Pharmaceutics* 13 (2021).